# 8*10/100/1000Base-T, 8*100/1000Base-X SFP

# 4*1G/2.5G/10GBase-X SFP+

# L3 Managed Industrial Ethernet Switch with Temp-humidity Sensor

**LT-IS6520XG-4XS8GX8GT-SD**



**Features:**

- 8*10/100/1000Base-T RJ45, 8*100/1000Base-X SFP, 4*1G/2.5G/10GBase-X SFP+
- Redundant power 12-75V input with polarity reverse/over-current protection
- Support DI/DO/RS485, work with temperature-humidity sensor to collect environment data and trigger alarms
- Support USB one key configuration & upgrading
- Layer 3 features: Support IPv4/IPv6 static routing, DHCP server, DHCP relay; Support ECMP, VRRP, RIP, OSPF, BGP4, IS-IS; 13K routing entries
- Layer 2 features: VLAN/GVRP/QinQ/SPAN/RSPAN/STP, RSTP, MSTP/DHCP/Multicast/ACL QoS/LLDP/802.1X/AAA/Loop Detect/Dying Gasp/SFP DDM/IPV6 management/Dual Stack/ sFlow/Web/SNMP/MQTT/Telnet/TFTP/Web upgrading
- Support G.8032 ERPS protocol, recovery time ≤20ms
- Support 6KV surge protection and ESD: Air-15kV, Contact-8kV Protection
- IP40 fan-less and Din-rail hardware design
- Operation temperature: -40 ℃ ~+75℃

## Overview

The SmartByte LT-IS6520XG-4XS8GX8GT-SD is the highly reliable managed industrial Ethernet switch with 8-port 10/100/1000-T, 8-port 100/1000Base-X SFP and 4-port 1G/2.5G/10GBase-X fiber optical interfaces. In normal operation, it is a L3 managed Ethernet switch. It supports ERPS redundant network and the self-recovery mechanism less than 20ms on full load which allows you to scheme a reliable Ethernet network by building a redundant ring topology as your back-up solution. It also supports DI/DO/RS485, DHCP server, DHCP relay, IPv4/IPv6 static routing and dynamic routing protocols like RIP, OSPFv2, BGP4, IS-IS, robust Layer 2 management and secure functions, which can help users expand their networks quickly and efficiently.

LT-IS6520XG-4XS8GX8GT-SD is the high cost-effective easy-to-use device, which provides essential industrial Ethernet networking function, including wide range power input 12-75VDC, redundant power design with polarity reverse protection, robust IP40 fan-less housing with Din-rail installation, wide operation temperature from -40°C to 75°C as well as high-level EMI/EMC capability. It is the best choice for heavy industrial factory, transportation, oil & gas, chemical, IP Surveillance and processing automation area where environmental conditions are harsh and critical.

## Technical specification

| | | |
|---|---|---|
| **Model No.** | LT-IS6520XG-4XS8GX8GT-SD | |
| **Interface** | Fiber ports | Copper RJ45 ports |
| | 12 | 8 |
| **Ethernet** | 8*10/100/1000Base-T RJ45<br>8*100/1000Base-X SFP (9-16 ports)<br>4*1G/2.5G/10GBase-X SFP+ (17-20 ports) | |
| **CPU** | Marvell AC5 series 1.2GHz | |
| **Size of RAM** | 1GB | |
| **Size of Flash** | 512MB | |
| **Switch chip model** | Marvell AC5 series | |
| **Management port** | 1*RJ45 Console port (115200,8,N,1)，1*USB 2.0 | |
| **Data & Alarm channel**（Removable 8-pin terminal block) | Vout: VO + , VO - | 5V output（0.2A MAX） |
| | 1 channel RS485: A, B | Work with temperature-humidity sensor to collect environmental data and trigger alarms |
| | 1 channel digital input: DI+，DI- | Status “1”: 12V-30V<br>Status”0”: -30V~3V<br>Max input current: 8 mA |

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| | 1 channel digital output | Normally open , Normally closed , Pulse<br>2A@30V DC，0.3A@125V AC |
| **Standard** | IEEE 802.3 10Base-T Ethernet<br>IEEE 802.3u 100Base-TX Ethernet<br>IEEE 802.3ab 1000Base-T Ethernet<br>IEEE 802.3z 1000Base-X Ethernet<br>IEEE 802.3ae 10Gigabit Ethernet<br>IEEE802.3x Flow Control and Back Pressure<br>IEEE 802.1D Spanning Tree Protocol<br>IEEE 802.1w Rapid Spanning Tree Protocol<br>IEEE 802.1Q VLAN<br>ITU-T G.8032 ERPS<br>IEEE 802.1X Port Authentication Network Control<br>IEEE 802.1ab LLDP<br>IEEE 802.3ad LACP | |
| **LED Indicators** | | |
| **P1/P2** (Power indicator) | Off: the device is power off or failed | On: the device power on is normal |
| **SYS (**System status indicator) | Blinking: device initialization | On: device on normal operation |
| **USB (**USB status indicator) Green | Off: USB flash drive not insert or not detected by device | |
| | On: USB flash drive is detected by device | |
| | Blinking: the device is reading and writing the data from USB flash drive | |
| **ALM** (Fault status indicator) | Off: Device on normal working state | On: Fault alarm |
| **Ring** (Ring network indicator) | Off: Ring network disabled | On: Ring network enabled |
| **1-8** (Copper ports indicators) Green | Off: ports link down | |
| | On: ports link up | |
| | Blinking: data on TX/RX | |
| **1-8** (Port speed indicators) Yellow | Off: ports speed on 10/100M | |
| | On: ports speed on 1000M | |
| **9-20** (Fiber ports indicators) Green | Off: ports link down | |
| | On: ports link up | |
| | Blinking: data on TX/RX | |
| **Power parameter** | | |
| Input voltage | 12-75VDC<br>redundant power input | |
| Input current | 1.8A Max | |

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| Power consumption | Full loading ≤22W |
| Connector | Removable 2*2 pin terminal block |
| Reverse polarity protection | Support |
| Over-current protection | Support |
| **Layer 2 switching** | |
| Port aggregation | Support GE port aggregation |
| | Support 2.5GE aggregation |
| | Support 10GE aggregation |
| | Support static aggregation |
| | Support LACP dynamic aggregation |
| | Up to 64 aggregation groups and up to 8 ports per group |
| Port features | Support IEEE802.3x flow control |
| | Support interface counters |
| | Support port isolation |
| | Port mirroring (One-to-One, Many-to-One) |
| | Support loop detection (Port-based; VLAN-based) |
| | Support broadcast storm suppression (broadcast; unknown multicast; unknown unicast) |
| VLAN | IEEE802.1Q tagged VLAN |
| | Mac based VLAN |
| | IP based VLAN |
| | Protocol based VLAN |
| | Voice VLAN, specially divide VLAN for user voice data flow |
| | Private VLAN, divide a VLAN broadcast domain into multiple subdomains to form a double-layer VLAN structure |
| | VLAN range: 1-4094 |
| | Support access mode/trunk mode/hybrid mode |
| Mac-address table management | Support static Mac-address management |
| | Support dynamic Mac-address Management |
| | Support filtering Mac-address |
| | Support MAC limit based on port and VLAN |

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| | Support MAC flapping based on port and VLAN |
| | 16K Mac-address table size |
| GVRP | Normal mode |
| | Fixed mode |
| | Forbidden mode |
| QinQ | Port-based QinQ |
| | VLAN-based QinQ |
| | VLAN mapping |
| | Flow-based QinQ |
| LLDP | Link Layer Discovery Protocol |
| | Med |
| SPAN | Number of Sessions: 7 |
| | Many-to-one |
| RSPAN | Remote Switch Port Analyzer |
| MSTP | 802.1D STP |
| | 802.1W RSTP |
| | 802.1S MSTP |
| | BPDU Guard |
| | Number of Single Ring Device Connections: 20 |
| ERPS G.8032 | Major ring |
| | Sub ring |
| | Recovery time ≤20ms |
| | Number of Single Ring Device Connections: No limit |
| QoS | QoS Class, Remarking |
| | Support SP, WRR Queue Scheduling |
| | Ingress Port-based Rate-limit |
| | Egress Port-based Rate-limit |
| | Egress port-queue shaper |
| | Policy-based QoS |
| | Port Queue Numbers: 8 |

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| DHCPv4 Snooping | DHCP snooping trust port |
| | DHCP snooping option-82 |
| | DHCP snooping circuit-id and remote-id customized |
| **L2 Multicast** | |
| IGMP | IGMP v1, v2, v3 |
| IGMP Snooping | Internet Group Management Protocol Snooping |
| | Fast Leave |
| | Multicast Table Size: 1K |
| PIM-SM | Protocol Independent Multicast-Sparse Mode |
| **Security features** | |
| ACL | IP Standard ACL<br>IP Extended ACL<br>MAC extend ACL<br>IPv6 ACL<br>Time Range<br>ACL apply on VLAN<br>ACL entries, include input and output |
| 802.1x | Port access control<br>Mac-address access control<br>RADIUS server |
| Port-security | Port-security |
| IP source guard | IP Source Guard, Port-based IP/Mac Binding Access |
| ARP-check | Support ARP-check and ARP packet filtering for illegal users |
| DAI | Dynamic ARP Inspection |
| Anti-gateway ARP Spoofing | Prevent Gateway ARP Spoofing |
| Ddos | Ddos attack protection |
| Access control | Support Telnet/SSH/HTTP/HTTPS user access control |
| AAA | RADIUS authentication access switch |
| | TACACS authentication access switch |
| | Authorization |
| **Reliability** | |
| Loop Detect | Loop detect, based on port or VLAN |
| EFM | The Ethernet in the first mile, 802.3ah |

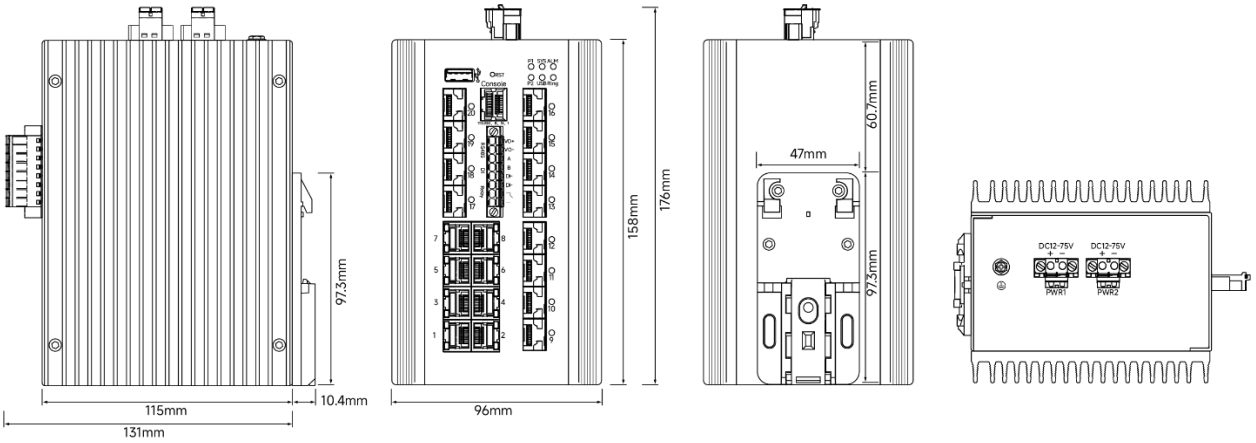
| **Management & Maintenance functions** | |
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| Dual Stack | IPv4 and IPv6 Dual Stack Management |
| System Log Management | Syslog Level 7 to 0<br>Corresponding Debug/Info/Notice/Warning/ Err/ Critical/ Alert |
| | Support connecting up to 3 Servers |
| | For IPv4 and IPv6 |
| User Management | Support password protection, support MD5/SHA256/SHA512 algorithm |
| | Local Management of User Privilege |
| | Remote authorization by RADIUS or TARCAS |
| | Enable privilege protection |
| | User password complexity requirements |
| | User login fail protection |
| | User Privilege Level 0 to 15 |
| Login Service Management | Up to 8 users online simultaneously |
| | Access control based on IPv4 or IPv6 ACL |
| Firmware Management | Firmware upgraded by WEB |
| | Firmware upgraded by CLI |
| | Dual partition |
| Web Management | Web Management |
| | Support HTTP V1.1 |
| | Support HTTPS |
| | For IPv4 and IPv6 |
| CLI management | Console/Telnet command line management |
| Fault Detection | Ping/Traceroute, For IPv4 and IPv6 |
| | Dying gasp (the input voltage should be more than 18V) |
| | Optical transceiver DDM |
| | One click fault information collection |
| | Cable detection |
| DNS | Domain Name System |
| NTP | Network Time Protocol, For IPv4 and IPv6 |

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| | Support domain name configuration |
| Configuration Import /Export | FTP/TFTP based remote configuration import and export |
| | USB flash drives based local configuration import and export |
| sFlow | Sampled flow |
| SNMP | Version SNMPV2C/V3 |
| MQTT | Message Queuing Telemetry Transport |
| RMON | Support RMON (Remote Monitoring) alarm |
| Dual partition management | Support dual partition switching |
| **Application Protocol Functions** | |
| Telnet server | Support telnet server for IPv4 and IPv6 |
| Telnet client | Support telnet client for IPv4 and IPv6 |
| SCP (Secure Copy Protocol) | SCP client/server for IPv4 and IPv6 |
| SFTP | sftp client/server for IPv4 and IPv6 |
| SSH server | Support login based on user password-SSH2, support IPv4, IPv6 |
| | Support public key-based login |
| TACACS | Support TACACS (the terminal access controller accesses the control system) |
| TFTP | Support TFTP Client |
| **Layer 3 features** | |
| Static Route | Support VLAN interface and route port |
| | IPv4 static routing table entry configuration |
| | ARP dynamic learning and static configuration |
| | IPv6 static routing table entry configuration |
| | Neighbor discovery and static configuration |
| | Routing prefix table size 4k for IPv4, for IPv6 divide by 4 |
| DHCPv4 Client | Dynamic host configuration protocol for IPv4 |
| DHCPv6 Client | Dynamic host configuration protocol for IPv6 |
| DHCPv4 Relay | DHCP Relay for IPv4 |
| DHCPv4 Server | DHCP Server for IPv4 |
| ECMP | Support ECMP (Equal-cost multi-path routing) |
| | Support the configuration of ECMP Max next-hop |

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| | Support capacity balanced configuration |
| Route policy | IPv4 prefix-list |
| VRRP | Virtual Router Redundancy Protocol |
| **Routing Protocol Specification** | |
| RIP | RIPv1/v2 |
| OSPFv2 | OSPFv2 |
| BGP4 | BGP4 |
| | BGP Support Routing Recursive ECMP |
| | Support to view the number of neighbors and up/down state |
| IS-IS | IS-ISv4 |
| Routing entries | 13K |
| **Switching features** | |
| Switching capacity | 112Gbps |
| Packet forwarding rate | 166.6Mpps |
| MAC address table | 16K |
| VLAN | 4K |
| Buffer | 12Mbit |
| Forwarding delay | <5us |
| Jumbo Frame | Support 10Kbytes |
| MDI/MDIX | Support |
| Watchdog | Support |
| **Mechanical structure** | |
| Case protection | IP40 |
| Installation method | Din-rail |
| Dimension(W*D*H) mm | 96*115*158mm |
| Weight | 1.90 kg |
| **Operating environment** | |
| Operating temperature | -40℃~+75℃ |
| Storage/transportation temperature | -40℃~+85℃ |

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| Relative humidity | Operation humidity: 10%-90%RH<br>Storage humidity: 5%-95%RH |
| **Industrial Standard** | |
| EMS | Surge protection of power: IEC 61000-4-5　6KV/4KV（8/20us） |
| | Surge protection of Ethernet ports: IEC 61000-4-5　6KV/2KV（10/700us） |
| | RS：IEC 61000-4-3 80 MHz-1 GHz：10 V/m |
| | EFT：IEC 61000-4-4 4K/2K |
| | CS：IEC 61000-4-6 10V |
| | ESD：IEC 61000-4-2　Contact: 8K；Air: 15K |
| EMI | FCC Part 15B Class A |
| Shock | IEC 60068-2-27 |
| Free fall | IEC 60068-2-32 |
| Vibration | IEC 60068-2-6 |
| **Certification** | CCC/CE/UKCA/RoHS/REACH/FCC |
| **MTBF** | > 200,000 hours |
| **Warranty** | 5 years |

## Structure diagram




## Order information

| Model | Description |
|---|---|
| LT-IS6520XG-4XS8GX8GT-SD | L3 Managed industrial Ethernet switch with 8*10/100/1000Base-T RJ45 ports , 8*100/1000Base-X SFP and 4*1G/2.5G/10GBase-X SFP+ slots, supports DI/DO/RS485 and temperature-humidity sensor on default, DC12-75V input, redundant dual power supply; Din-rail installation; Operation temperature: -40℃～+75℃ |